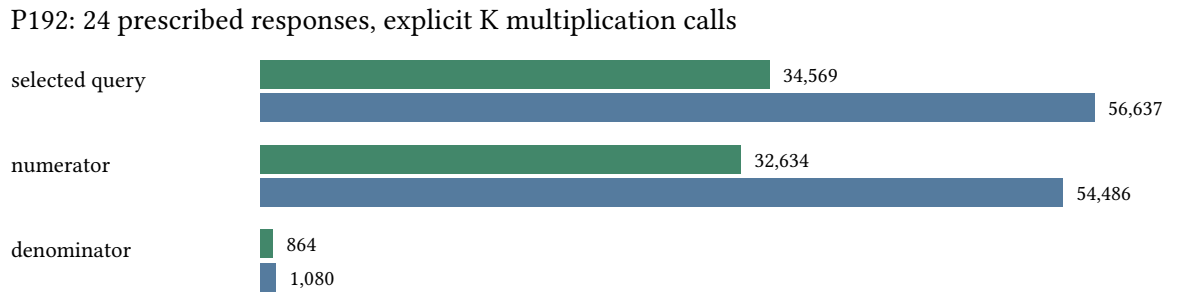
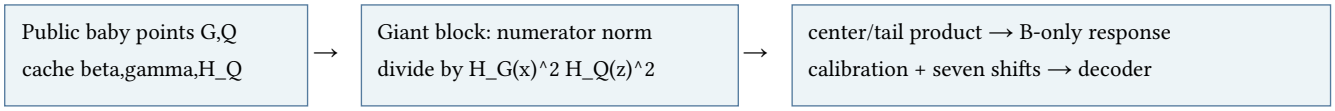
Reduced paired numerator by biquadratic cancellation



Green reduced; blue expanded. Shared scale. Setup separate. Source: audit-summary.json and raw P192 rows.



P192 executes prescribed 27-factor windows; the complete decoder panel runs on the small curves.

Round67, 2026-11-09, T-1983. Derived and implemented a common-factor cancellation in the signed pair numerator: E0=D*M0,E1=D*M1. The selected complete-block jet is now Norm(M0+s*M1)/(H_G(x)^2*H_Q(z)^2), replacing the expanded numerator over the fourth-power denominator. The independent coefficient certificate and frozen response controls pass. Scientific standing remains unverified producer evidence. [pcc:unverified]

Written construction. Let f(X)=X^3+aX+b, N(x,t)=t*x^2+(a+t^2)*x+a*t+2b, F(x,t)=(x*t-a)^2-4b*(x+t). The classical identity N(x,t)^2-4f(x)f(t)=(x-t)^2F(x,t) supplies the cancellation. For the unchanged baby algebra put D_G=(x-T)^2,D_Q=(z-beta)^2,D=D_G*D_Q and L=N_G*D_Q-N_Q*D_G. Then M0=F_G*D_Q+F_Q*D_G-2*N_G*N_Q+8*y*w*gamma, M1=2L. The denominator is a unit by the existing nonrational-shift guards, so the singular constant-numerator cases remain valid. Full derivation and the consequent coefficient-cache design are in cancellation-proof.md. [pcc:unverified]

Universal coefficient check. A separate native bin uses sparse polynomials over Z[x,t,z,v,a,b,c], with c=y*w*gamma. It uses checked i128 arithmetic and exponent addition, no elliptic or finite-field helper. Direct expansion checks the biquadratic identity, E0=D*M0 and E1=D*M1, retaining both sides and all coefficients. E0 has251 nonzero terms; the correct residual is zero. Changing the gamma sign produces9 residual terms. This checks the algebra before substituting v=beta(T), independently of the finite-curve controls. [pcc:unverified]

Frozen response checks. All381 calibrated decoder cases across2694 coefficient queries preserve the previous scientific output. All11382 complete small-curve baby blocks match direct pairs. The expanded control's scientific output, cost receipt and P192 output are byte-identical to sealed Round66 candidate records, confirming the control has not changed. The candidate's small scientific output is byte-identical to that control. Original point inputs, cache constructors, relative-sign guards, clipped tails and decoder remain. [pcc:unverified]

Primary P192. All24 prescribed27-factor responses and72 degree4 rational baby norms match the earlier direct/reference coefficients on publicQ,mu_G,relation_lambda_G. All54 unit and18 simple-common-root numerator cases persist. Selected query totals are34569 multiplication calls,1461 inversions and168 point-add calls versus56637,1605,168 for the expanded numerator. Numerator formation/norm uses32634 multiplications/1152 inversions versus54486/1296; denominator work864/0 versus1080/0; output divisions144/144 in both. P192 counts exclude inverse internals. This is the prescribed-window scope; no complete order-r P192 response is executed. [pcc:unverified]

Small integrated selected-query totals (multiplications including inverse powers,inversions,point-add calls) are: r7 reduced(99404,4274,1400) versus expanded(133036,5976,1400); r19(3295513,123267,19296) versus(4535925,168425,19296); r199(12338608,261615,10150) versus(19264532,384488,10150). Direct complete-block validation is separately charged and subtracted from selected-query counters; whole-qualifier records retain all reference work. Source/target setup and decoding are separate. No timings or cross-counter-model ratio are inferred. [pcc:unverified]

Setup and validation. Target construction is unchanged: each P192 H_Q cache costs20 multiplications and5 coefficients; the qualifier retains3 such caches. Beta/gamma/H_Q need13 polynomial coefficient slots per target, source H_G another5; temporary/reference memory is additional. Scope and native source37dfafaac2bd30bc7642c09bc8ad6a318240332f were published before data. Five science outputs and two operation receipts match a fresh-copy build. Altered calibration and P192 coefficient controls reject101. Build succeeded with one unused-helper warning, retained in build.log. The old library compilation timeout124 is inherited without retry. No cloud job, generator, role, quota or old freeze lane changed. [pcc:unverified]

Next construction. After isolating8*y*w*gamma, M0 and M1 have degree at most2 in each giant x-coordinate. Cache their9+9 coefficient polynomials modulo H_G after the target beta is fixed. This gives a concrete path to O(b) numerator formation per giant block via scalar linear combinations, with at most18*b extra field coefficients and explicit setup cost. This round derives that representation; the successor must implement and independently check the cache before assigning measured costs. [pcc:unverified]

Sources checked. Round63 paired-algebra-proof.md, Round64 resultant-proof.md and Round66 report/continuation; Bernstein-De Feo-Leroux-Smith, https://arxiv.org/pdf/2003.10118, Lemma4.3 and section4.16, read2026-11-09. Classical biquadratics motivate the simplification; no novelty claim for that identity or imported fast coupled-resultant bound. KB tools/wrapper unavailable. [pcc:unverified]